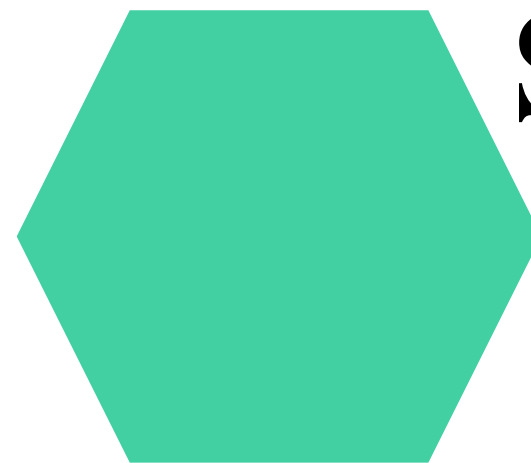




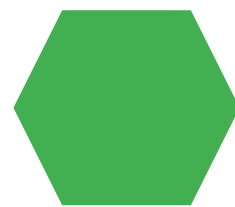
Seasonal Agriculture Performance Analysis

SUBMITTED BY :-

Yash Tolani

College :-

Indore Institute of Science and Technology



AICTE Student ID :-

STU682f3bce808201747925966

PROBLEM STATEMENT

- **Context :-** Agricultural performance varies significantly across seasons due to changing environmental conditions, irrigation practices, and market prices.
- **Core Issue :-** Raw agricultural data is complex and fails to directly explain seasonal trends, resource efficiencies, or profitability variations.
- **Objective :-** Analyze 4,000 farm records across 28 parameters to identify seasonal patterns, yield differences, resource utilization, and economic outcomes.



PROJECT DESCRIPTION

- **Scope :-** End-to-end Data Analytics project analyzing agricultural performance across Kharif, Rabi, and Zaid seasons in 8 major Indian states.
- **Data Processing :-** Cleaned missing values (Rainfall, Soil Moisture, Yield) and engineered key performance indicators like Profit Margin %.
- **Analytical Approach :-** Exploratory Data Analysis (EDA), comparative seasonal profiling, correlation analysis, and statistical hypothesis testing.
- **Key Focus Areas :-** Crop yield evaluation, water efficiency, disease/pest risk assessment, and financial viability (Revenue vs. Cost).

WHO ARE THE END USERS ?



- **Farmers & Agricultural Advisors :-** Helps select optimal crop-season combinations to maximize profitability and minimize loss.
- **Agricultural Policy Makers :-** Supports evidence-based decisions for seasonal resource distribution, fertilizer allocations, and subsidies.
- **Irrigation & Water Resource Managers :-** Assists in optimizing irrigation methods (Drip vs. Flood/Sprinkler) during water-scarce seasons like Zaid.
- **Agri-Fintech & Supply Chain Analysts :-** Aids in risk modeling, disease risk tracking, and market price forecasting.



TECHNOLOGY USED

- **Programming Language :-** Python 3 – Core language for data processing, EDA, and statistical modeling.
- **Data Manipulation :-** Pandas & NumPy – Data cleaning (120 missing values), grouping, and feature engineering (Profit Margin %).
- **Data Visualization:** Matplotlib & Seaborn – Generating Heatmaps, Seasonal Profit Boxplots, and Bar Charts.
- **Statistical Analysis:** SciPy (scipy.stats) – One-Way ANOVA testing ($p\text{-value} = 1.71 \times 10^{-15}$) and Pearson correlation analysis.
- **Environment:** Google Colab / Jupyter Notebook – Interactive IDE for code execution and visual reporting.



RESULTS [Seasonal & State-wise Yield Performance]

16.5. Best State

```
▶ best_state = state_yield.idxmax()
  best_state_yield = state_yield.max()

  print("Highest average yield state:", best_state)
  print("Average yield:", round(best_state_yield, 2), "Tonnes/Ha")
```

```
... Highest average yield state: Punjab
  Average yield: 6.12 Tonnes/Ha
```

16.3. Best Crop

```
best_crop = crop_yield.idxmax()
best_crop_yield = crop_yield.max()

print("Highest yielding crop:", best_crop)
print("Average yield:", round(best_crop_yield, 2), "Tonnes/Ha")
```

```
Highest yielding crop: Sugarcane
Average yield: 46.94 Tonnes/Ha
```

RESULTS [Crop Economics & Market Price Trends]

16.1. Best Season

```
best_season = season_yield.idxmax()
best_yield = season_yield.max()

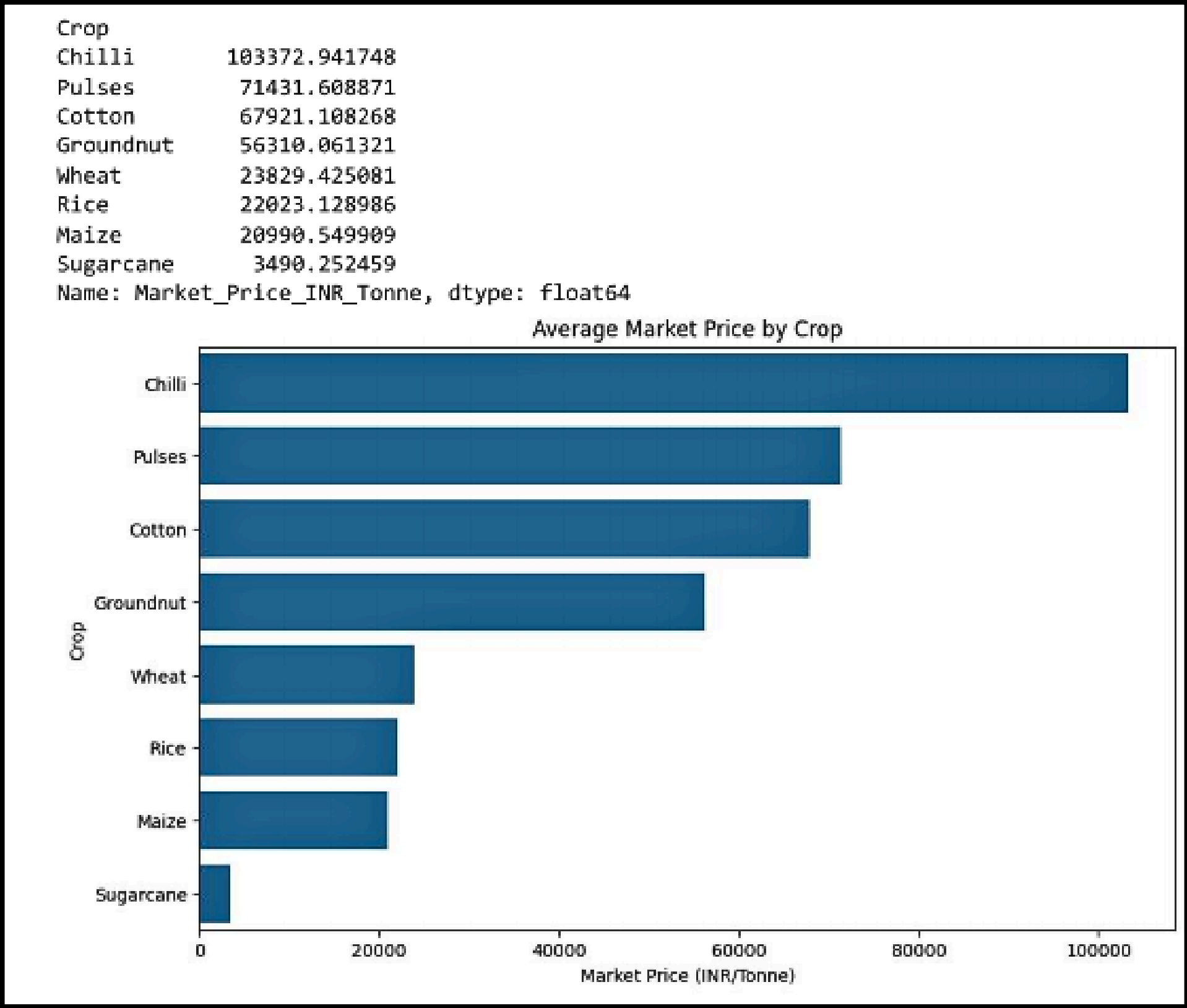
print("Best performing season:", best_season)
print("Average yield:", round(best_yield, 2), "Tonnes/Ha")
```

Best performing season: Kharif
Average yield: 5.64 Tonnes/Ha

	Yield_Tonnes_Ha	Production_Tonnes	Revenue_INR	Total_Cost_INR	Profit_INR	Water_Used_m3
Crop						
Sugarcane	46.94	392.41	1371980.11	554792.13	817187.99	13897.70
Chilli	1.54	12.38	1276777.23	525898.89	750878.34	4577.47
Cotton	1.23	9.40	639423.08	514876.17	124546.92	5410.86
Groundnut	1.32	9.66	542935.50	498077.38	44858.12	3459.54
Pulses	0.92	7.40	528465.87	532703.92	-4238.05	2833.67
Maize	2.72	21.78	457067.99	541046.32	-83978.33	4377.78
Rice	2.44	19.29	423615.26	525828.76	-102213.50	10920.22
Wheat	2.11	16.85	400104.89	523503.22	-123398.34	4054.40

RESULTS

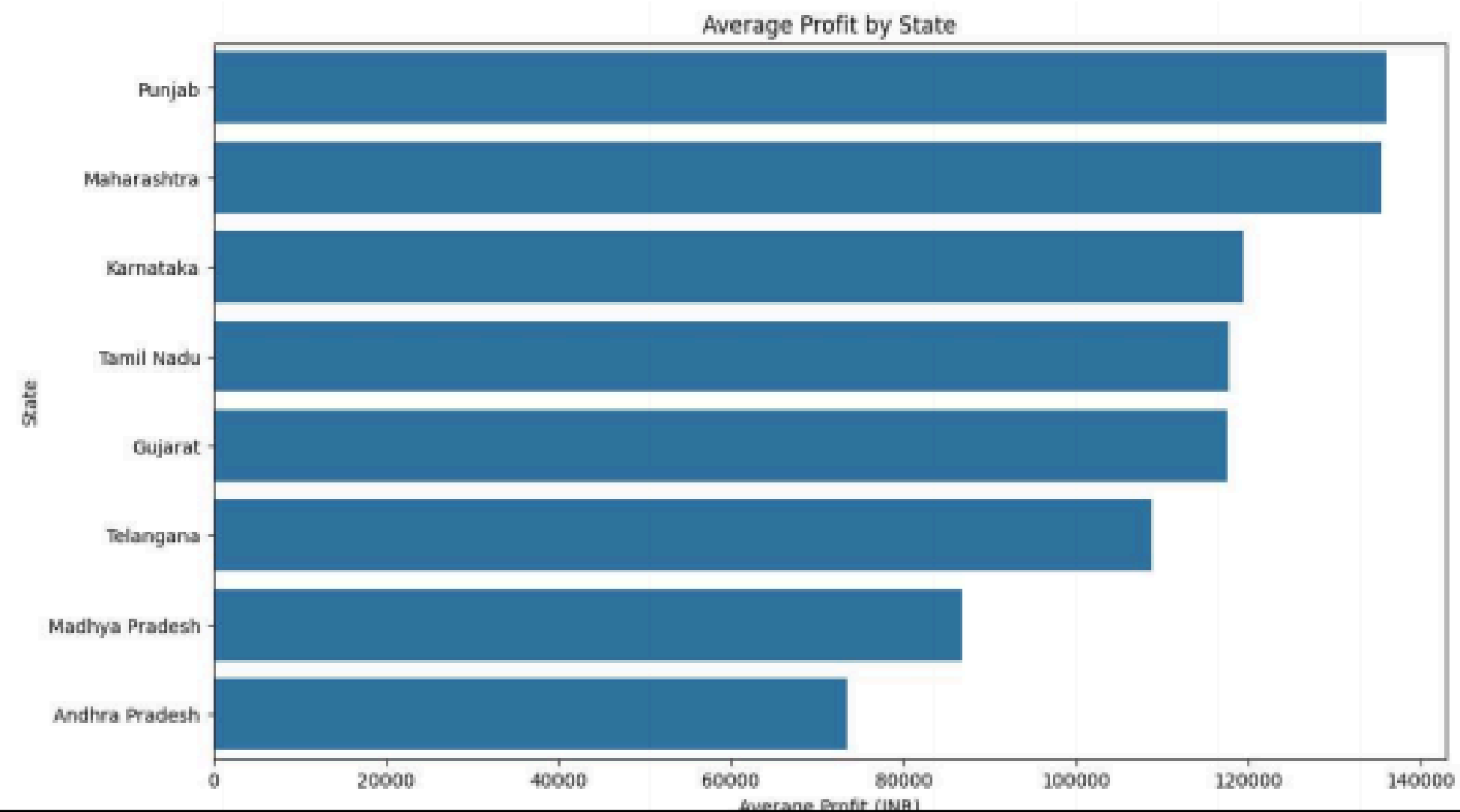
[Regional Revenue & Cost Breakdown]



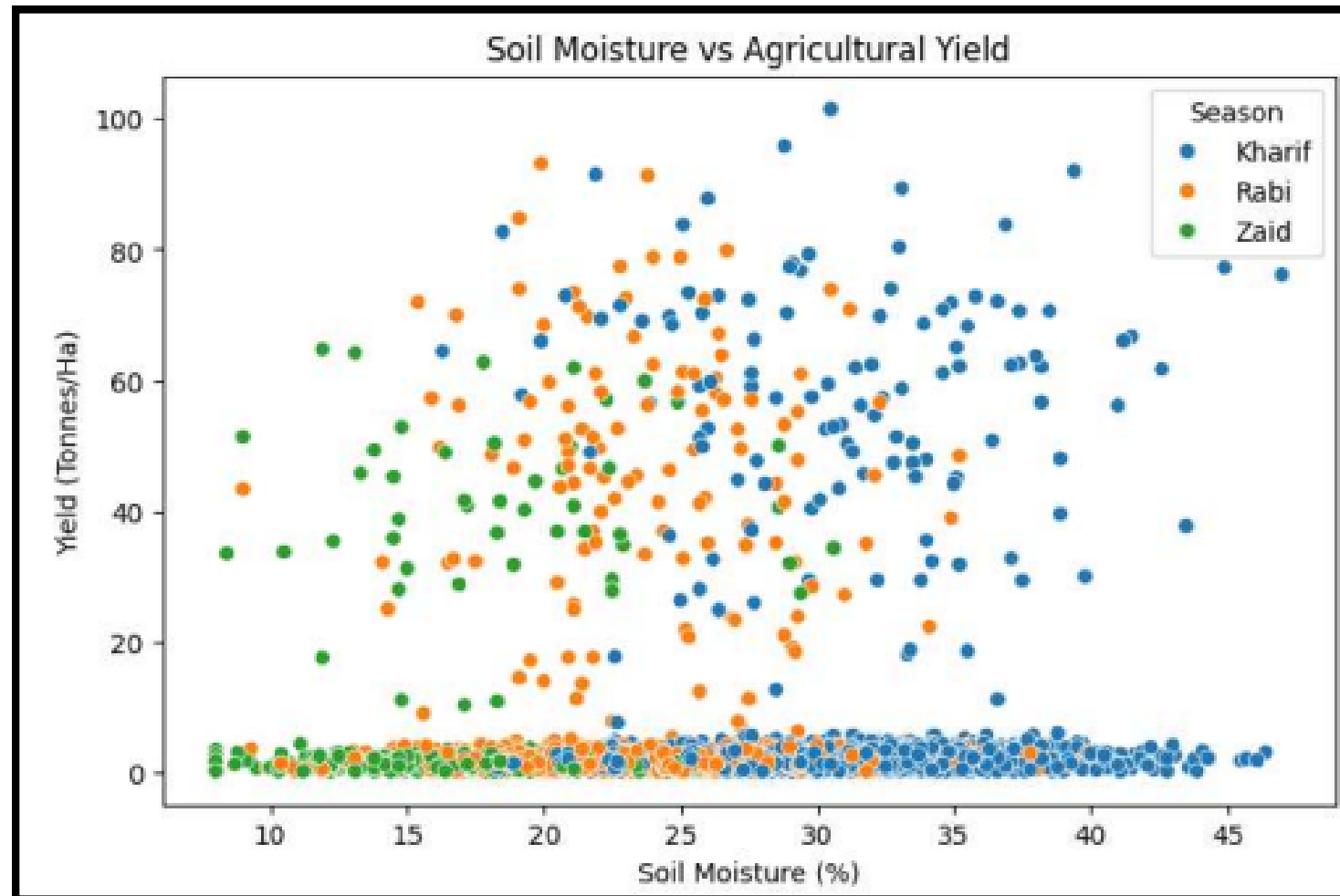
RESULTS

[Average Profit By State]

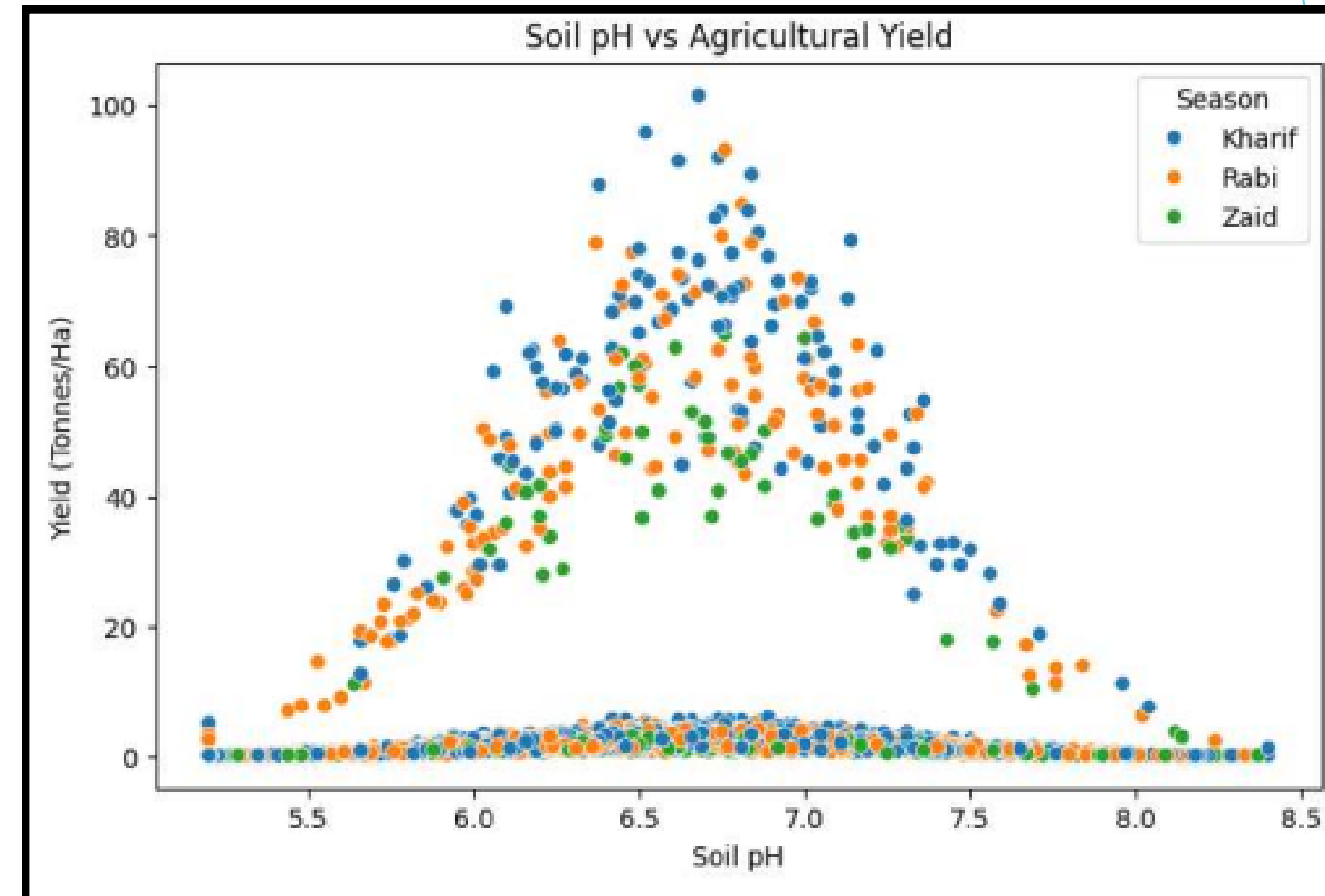
```
State
Punjab      136025.636364
Maharashtra 135428.855469
Karnataka   119422.335378
Tamil Nadu  117744.748454
Gujarat     117568.235656
Telangana   108829.616279
Madhya Pradesh 86840.756539
Andhra Pradesh 73453.527410
Name: Profit_INR, dtype: float64
```



RESULTS [Soil pH & Moisture Impact on Yield]



Soil Moisture vs Agricultural Yield



Soil pH vs Agricultural Yield

- **Soil pH :-** Optimal crop yield (up to 100 Tonnes/Ha) is achieved within a neutral pH range (6.5-7.0). Yield declines sharply when pH falls below 6.0 or exceeds 7.5.
- **Soil Moisture :-** Peak yield is concentrated between 20%-35% soil moisture. Moisture levels below 15% (characteristic of the Zaid season) severely restrict crop yield.

FUTURE SCOPE

- **Machine Learning Integration :-** Build predictive models (e.g., Random Forest / XGBoost) for pre-planting yield and price forecasting.
- **Interactive Dashboard :-** Develop a Streamlit or Power BI dashboard for real-time decision-making for farmers and extension officers.
- **Real-time Weather & Sensor Data :-** Integrate live IoT weather and soil moisture sensors for precision farming alerts.
- **Smart Crop Recommendation Engine :-** Implement AI algorithms to recommend crop rotations based on water availability and profit margins.

GITHUB LINK

GitHub Repository Link

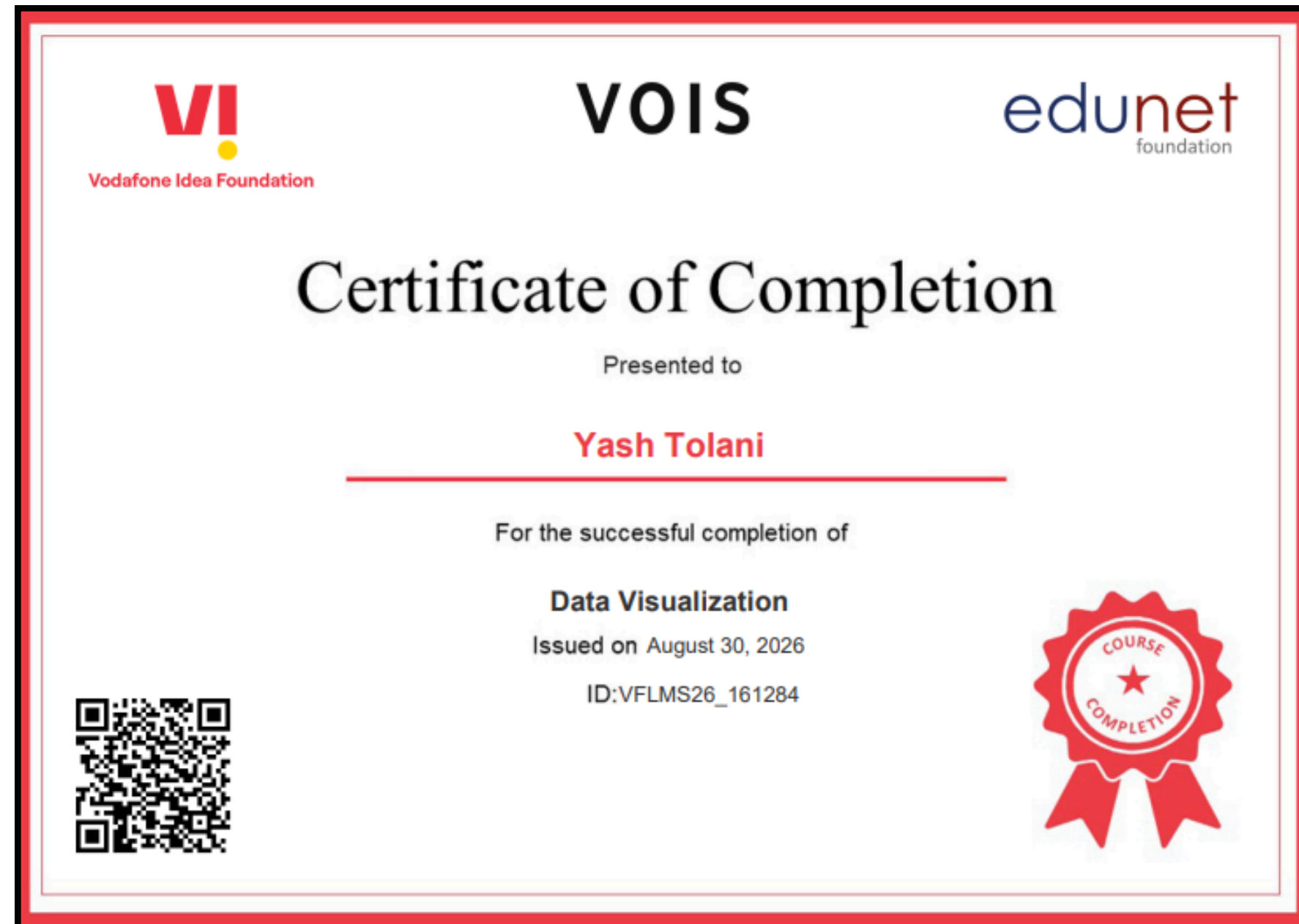
[<https://github.com/Yash192004/Seasonal-Agriculture-Performance-Analysis-Project>]

Repository Contents & Deliverables :-

- **Dataset (seasonal_agriculture_performance_dataset.csv) :-** 4,000 raw agricultural farm records covering 28 environmental, financial, and operational variables.
- **Project Statement (Project_Statement.pdf) :-** Project scope, objective guidelines, problem definition, and key analytical questions.
- **Colab Notebook Code (Seasonal_Agriculture_Performance_Analysis.ipynb) :-** Complete Python source code for data cleaning, EDA, ANOVA hypothesis testing, and Seaborn visualizations.
- **Presentation Deck (Seasonal_Agriculture_Performance_PPT.pptx) :-** Major project submission PPT covering data workflow, visual results, state/season analysis, and future recommendations.
- **Documentation (README.md) :-** Comprehensive project documentation, execution guide, methodology summary, and core insights.

COURSE CERTIFICATION

- **Course Name :-** Data Visualization
- **Issuing Authority :-** VOIS / VI Foundation / Edunet Foundation



**THANK
YOU**